

STAT395: Sampling

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```
library(haven)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(ggplot2)
library(kableExtra)
```

```
##
## Attaching package: 'kableExtra'

## The following object is masked from 'package:dplyr':
##
##   group_rows
```

```
library(survey)
```

```
## Loading required package: grid
## Loading required package: Matrix
```

```
## Loading required package: survival
##
## Attaching package: 'survey'
## The following object is masked from 'package:graphics':
##
##      dotchart
library(srvyr)

##
## Attaching package: 'srvyr'
## The following object is masked from 'package:kableExtra':
##
##      group_rows
## The following object is masked from 'package:stats':
##
##      filter
library(sampling)

##
## Attaching package: 'sampling'
## The following objects are masked from 'package:survival':
##
##      cluster, strata
#library(gtsummary) # error installing/loading this
library(mice)

##
## Attaching package: 'mice'
## The following object is masked from 'package:stats':
##
##      filter
## The following objects are masked from 'package:base':
##
##      cbind, rbind
#library(simputation) # error installing/loading this
```

Chapter 1

Introductory Remarks

These are notes for the Sampling section of STAT395/455. The course overlaps to some extent with STAT392/439, but there is a greater emphasis on R coding, computation, summary and display.

1.1 Lecture Plan

Course Learning Objectives

- Describe the steps in the survey process, including the sources of survey error;
- Draw samples from a sample frame under a variety of sample designs, and compute survey weights;
- Carry out, interpret and present statistical inference procedures on survey data, including adjustments for non-response.

Lecture plan (3 lectures + 1 tutorial per week)

Week 1 – The Survey Process

- The survey process, sources of survey error
- Populations and Frames
- Sampling in finite and infinite populations; Basics of SRSWOR inference; sample weights;
- **Tutorial:** Population and Frames

Week 2 – Sampling

Use of `sample()` and other simulation methods in R for a variety of designs [`sampler` package (does power calculations, allocation, SRS and STSRS, but not cluster)] [`sampling` package – can do cluster samples]

- Simple random sampling, with and without replacement
- Stratified sampling
- Cluster sampling
- Sampling in complex designs
- Sample size calculation and allocation
- Survey weights – create estimates from weights without standard errors
- **Tutorial:** using R to draw samples and compute survey weights

Week 3 - Inference with the R survey package

- Inference in simple and complex designs, including regression
- Variance estimation
- Definition and interpretation of the design effect
- **Tutorial:** using R to specify designs and draw inferences

1.1.1 Week 4 – Non-response and complex designs

- Post-stratification for non-response adjustment
- Imputation (`imputation` package)
- Jackknife and bootstrap inference
- **Tutorial:** raking adjustments to weights; imputation; jackknife inference

Assignment 4 (10%) [due end of Week 12: covers material from Weeks 9-11]

- Survey process, populations and frames
- Describe a real survey, comment on design
- Propose a sample design, frame, comment on scope and coverage
- Draw and analyse a sample under various designs
- Compute sample sizes and allocate samples in stratified sampling
- Compute and interpret the design effect
- Analyse data and present results from a survey

Test 2 (30%) [assessment period] [1.5 hours, closed book]

- Survey process, populations and frames
- Describe sample designs
- Compute and interpret sample weights
- Use sample weights for point estimates
- Interpret output from survey inference in R
- Compare results, interpret design effect
- Describe methods for non-response adjustment (imputation and post-stratification); comment on bias reduction and assumptions being made
- Describe the jackknife method of variance estimation

1.2 Recommended reading

There are lots of books on sample surveys.

- Thomas Lumley’s book ‘Complex Surveys’ (Lumley (2010)) is a good introduction to sample survey analysis with R, and accompanies the `survey` package (Lumley (2004)).
- Groves et al. ‘Survey Methodology’ (groves.etal) is a good general summary of the important issues in survey design.
- Sarndal, Swensson and Wretman is a classic in the field of the theory of sampling (Sarndal et al. (1992))
- Likewise the books by Cochran (1977) and Kish (1965)
- Little and Rubin have a great book on missing data (Little and Rubin (2002))
- More recent are the books by Lohr (1999) and Scheaffer et al. (2006)
- And in 1994 StatsNZ published ‘A Guide to Good Survey Design’ (Zealand (1995))

And also

- CRAN site on R resources for Official Statistics <https://cran.r-project.org/web/views/OfficialStatistics.html>

Some simple lectures on Sample Surveys

- https://r-project.ro/conference2017/presentations/Camelia_Goga_sondages.pdf
- Notes for epidemiologists: https://www.epirhandbook.com/en/new_pages/survey_analysis.html
- Notes on imputation: <https://www.r-bloggers.com/2023/01/imputation-in-r-top-3-ways-for-imputing-missing-data/>

Data

- UC Irvine Machine Learning data repository: <http://archive.ics.uci.edu/>
- Some free microdata sets: <https://asdfree.com/>
- Kaggle data repository: <https://www.kaggle.com/>

1.3 R Packages

- **survey** package (Lumley (2004)) with this handy summary of command uses <https://stats.oarc.ucla.edu/r/seminars/survey-data-analysis-with-r/>
- **srvyr** a dplyr wrapper for the **survey** package
- **gtsummary** creates outputs ready for publication (difficult to build/load)
- **sampling** package for drawing samples according to various designs
- **mice** package for imputation
- **simputation** package for imputation (difficult to build/load)
- **haven** package to import **SAS**, **Stata** and **SPSS** data files

Chapter 2

Sampling

The generation of data often includes the selection of a subset of members of a **population**. This subset is called a **sample**.

Our focus in statistics is often the **estimation** of population characteristics (**population parameters**) on the bases of sample characteristics (**sample statistics**),

When generating **simulated** datasets we also generate samples. Simulation is often done as a means of testing out statistical ideas and methods, or (in the case of the bootstrap method) when conventional ways of estimating standard errors (and therefore confidence intervals) don't work.

2.1 Random numbers

In both settings (sampling and simulation) we need to start with a method of generating random numbers. We'll start by assuming we have some means of generating a random sequence of numbers uniformly distributed in the interval $[0,1]$:

$$U_i \sim \text{Uniform}(0,1) \quad \text{for } i = 1, 2, 3, \dots$$

In R this is done using the `runif()` function. For example to generate 10 random draws from $\text{Uniform}(0,1)$:

```
runif(10)
```

```
## [1] 0.2875775 0.7883051 0.4089769 0.8830174 0.9404673 0.0455565 0.5281055  
## [8] 0.8924190 0.5514350 0.4566147
```

Another draw will a different set of random numbers:

```
runif(10)
```

```
## [1] 0.95683335 0.45333416 0.67757064 0.57263340 0.10292468 0.89982497  
## [7] 0.24608773 0.04205953 0.32792072 0.95450365
```

If we draw a histogram of 100 draws it looks a bit lumpy:

```
hist(runif(100), xlab="x", ylab="Frequency")
```

But a histogram of 100,000 draws is very uniform:

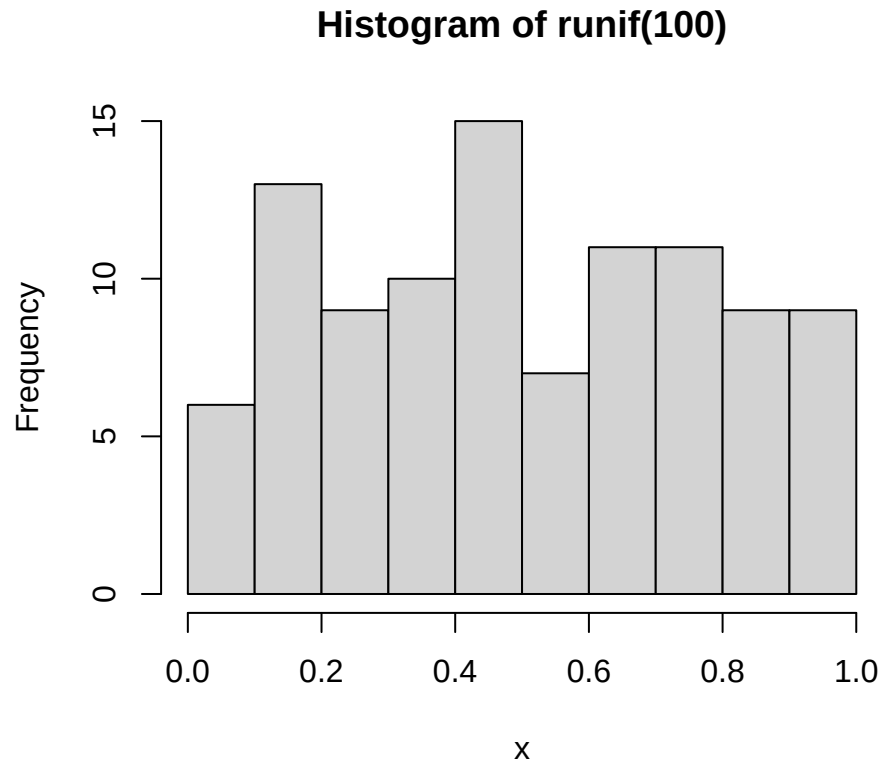


Figure 2.1: 100 draws from Uniform(0,1)

```
hist(runif(100000), xlab="x", ylab="Frequency")
```

Although the numbers appear to be random they are in fact generated using formulae, so if one knows the formula they are predictable. However the formula is complex enough that there are no easily discernible patterns in sequences of many millions of draws. These numbers are **pseudo-random**, but they are random enough for our purposes.

Underlying the random number generator is a **seed** value, which initiates the sequence. If we set the seed to a particular value, the sequence of random numbers that will be generated after doing so will always be the same.

```
set.seed(111)
runif(10)
```

```
## [1] 0.59298128 0.72648112 0.37042200 0.51492383 0.37766322 0.41833733
## [7] 0.01065785 0.53229524 0.43216062 0.09368152
```

```
runif(10)
```

```
## [1] 0.55577991 0.59022849 0.06714114 0.04754785 0.15620252 0.44642776
## [7] 0.17144369 0.96653429 0.31066643 0.61446640
```

```
runif(10)
```

```
## [1] 0.4310608 0.2855271 0.3421513 0.3866276 0.9675275 0.3220267 0.6532295
## [8] 0.2833035 0.7874279 0.5959206
```

```
set.seed(8)
runif(10)
```

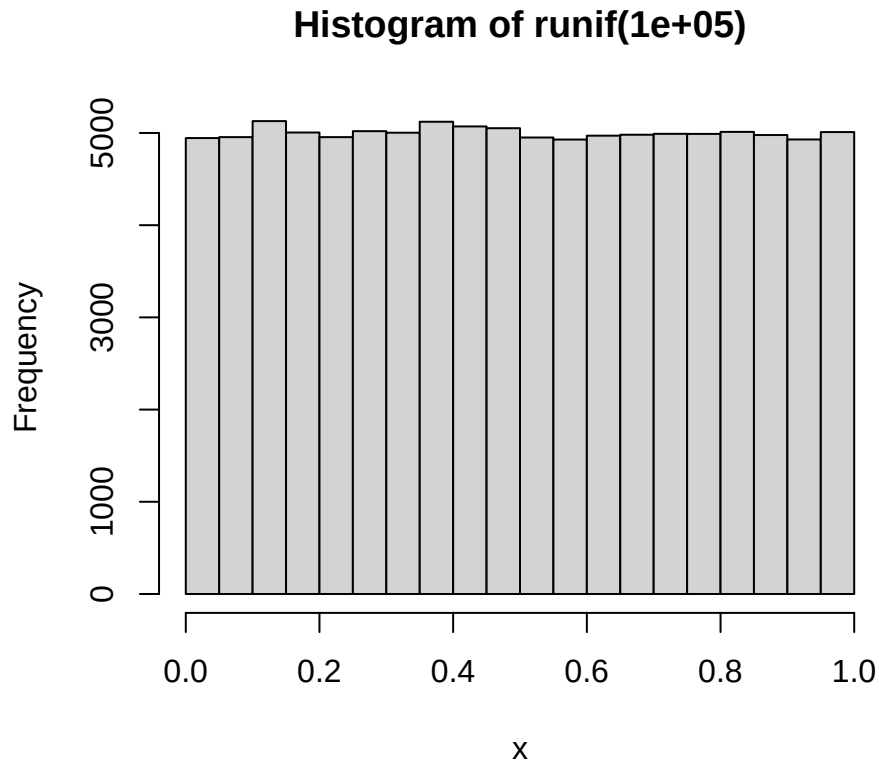


Figure 2.2: 100000 draws from Uniform(0,1)

```
## [1] 0.4662952 0.2078233 0.7996580 0.6518713 0.3215092 0.7189275 0.2908734
## [8] 0.9322698 0.7691470 0.6444911
```

```
set.seed(111)
runif(10)
```

```
## [1] 0.59298128 0.72648112 0.37042200 0.51492383 0.37766322 0.41833733
## [7] 0.01065785 0.53229524 0.43216062 0.09368152
```

This property can be very useful when testing algorithms.

Chapter 3

The Survey Process

Steps in survey sampling

3.1 Populations

3.2 Frames

References

Appendix A

Computing the Required Sample Size

1. Identify your **design estimate** – this is the estimate that will underlie a headline you could imagine being written about your survey.
2. Identify the desired MOE, m , for this design estimate, noting whether you want this MOE to apply to the overall estimate, or for that estimate in each of, say, H subgroups/strata.
3. Compute the sample size required, n_1 , assuming that you’re going to select a Simple Random Sample and ignoring the fpc.

For estimation of a mean $\bar{Y}$ of some characteristic y :

$$n_1 = \left(\frac{Z}{MOE} \right)^2 s^2$$

where Z is appropriate for the desired level of confidence and s is the estimated population standard deviation of the characteristic y .

For estimation of a proportion P :

$$n_1 = \left(\frac{Z}{MOE} \right)^2 p(1 - p)$$

where p is the estimated proportion, or 0.5 if really unknown.

4. If the sample size is a substantial proportion of the population or subgroup size N , then apply the fpc adjustment:

$$n_2 = \frac{n_1}{1 + \frac{n_1}{N}}$$

5. If you have H subgroups and are using equal allocation then multiply by H

$$n_3 = Hn_2$$

or alternatively sum over the n_2 values in each of the strata you’ve defined.

6. Apply the design effect (Deff) appropriate for the actual design you’ll be using.

$$n_4 = \text{Deff} \times n_3$$

7. Adjust for anticipated non-response: if response rate ϕ is expected then

$$n_5 = \frac{n_4}{\phi}$$

(If non-response is likely to vary between strata, then apply this correction to n_2 , within each stratum.)

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